

Optics Letters

Graphene electrodes for lithium-niobate electro-optic devices

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Received 1 February 2018; revised 10 March 2018; accepted 11 March 2018; posted 13 March 2018 (Doc. ID 321219); published 6 April 2018

We propose and demonstrate the use of graphene electrodes for lithium-niobate electro-optic (EO) devices to exempt the need of incorporating a buffer layer between the waveguide and the electrodes. Using graphene electrodes, our experimental mode converter, based on an EO-generated long-period grating in a LiNbO₃ waveguide, shows a reduction in the half- π voltage by almost three times, compared with the conventional electrode design using metal. With the buffer layer exempted, the device fabrication process is also significantly simplified. The use of graphene electrodes is an effective approach to enhancing the efficiency of EO devices and, at the same time, reducing their fabrication cost. © 2018 Optical Society of America

OCIS codes: (130.3120) Integrated optics devices; (130.3730) Lithium niobate; (130.4815) Optical switching devices; (310.3915) Metallic, opaque, and absorbing coatings.

<https://doi.org/10.1364/OL.43.001718>

Lithium-niobate (LiNbO₃) electro-optic (EO) devices, which operate on electrical modulation of the refractive index of a LiNbO₃ optical waveguide through the EO effect, have found numerous applications as optical modulators and switches in optical communication, optical signal processing, and optical sensing systems [1–4]. To exploit the largest EO coefficient r_{33} of the LiNbO₃ crystal, the optics axis (the z -axis) of the crystal, the applied electric field, and the major electric field component of the guided mode should be aligned. Therefore, a z -cut LiNbO₃ device should be designed for the transverse-magnetic (TM) mode, whose major electric-electric component is perpendicular to the waveguide surface, while an x -cut or a y -cut LiNbO₃ device should be designed for the transverse-electric mode, whose electric field component is parallel to the waveguide surface. To maximize the applied electric field experienced by the guided mode and, hence, minimize the driving voltage, the electrodes that generate the applied electric field should be placed as close to the waveguide core as possible. As conventional metal strongly absorbs light, a buffer layer of low-index dielectric, such as silicon dioxide (SiO₂), is usually placed between the waveguide core and the electrodes to reduce the metal-induced loss. The buffer layer, though very thin (no more

than a few hundred nanometers), may significantly reduce the EO effect experienced by the guided mode. Nevertheless, an electrode design that incorporates a buffer layer has been the established methodology in the development of EO devices for many decades [1–4]. In this Letter, we propose using graphene as the electrode material for EO devices, so that the electrodes can be placed directly on the surface of the waveguide, i.e., the buffer layer is exempted. In this way, we can truly optimize the applied electric field in the waveguide and, hence, minimize the driving voltage.

Our proposed idea is based on our recent study of the optical properties of graphene [5,6]. Being a single layer of carbon atoms, graphene has been widely used in optical devices, such as mode-locked lasers [7], polarizers [8], modulators [9], and sensors [10]. Our theoretical analysis and experimental results prove that graphene strongly attenuates the propagating wave whose major electric field is parallel to its surface, while introducing little loss to the propagating wave whose major electric field is perpendicular to its surface [5]. We have applied this property of graphene to the realization of mode-selective filters for mode-division-multiplexing applications [6]. According to this finding, for a z -cut LiNbO₃ device designed for the TM mode, electrodes formed with graphene can be placed directly on the waveguide without inducing significant loss to the guided mode. A significant reduction in the driving voltage of the device should be expected. Without the need of growing a dielectric buffer layer on the LiNbO₃ waveguide, the fabrication process also should be simplified. To demonstrate the feasibility of the proposed idea, we fabricated an EO mode converter with a z -cut LiNbO₃ waveguide, where a long-period grating is generated electrically with periodic electrodes to achieve conversion between two guided modes. The half- π voltage of our device is reduced almost three times by using graphene electrodes, compared with the conventional electrode design using aluminum (Al). Our EO mode converter could find applications as a dynamic wavelength filter or a mode switch.

Figures 1(a) and 1(b) show the structures of a conventional grating-based LiNbO₃ EO mode converter [11] and the proposed device using graphene electrodes, respectively. We assume a z -cut LiNbO₃ waveguide formed by the proton-exchange (PE) process. Since the PE process increases only the extra-ordinary refractive index of the LiNbO₃ crystal, the resultant waveguide supports only the TM-polarized modes. The grating, which is

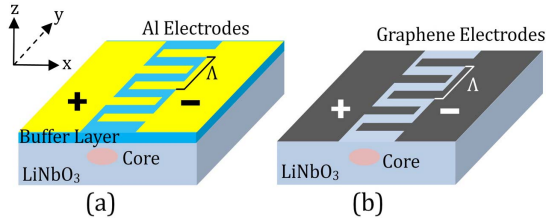


Fig. 1. Schematic diagrams of LiNbO₃ EO mode converters with (a) the conventional Al electrodes and (b) the graphene electrodes.

generated electrically by the periodic electrodes, is designed for coupling between the TM-polarized LP₀₁ and LP_{11b} modes. The resonance wavelength, at which the coupling effect is strongest, is determined by the phase-matching condition [11,12]:

$$\lambda_0 = (N_{01} - N_{11b})\Lambda, \quad (1)$$

where N_{01} and N_{11b} are the effective indices of the LP₀₁ and LP_{11b} modes, respectively, and Λ is the pitch of the grating. The strength of the mode-coupling effect is controlled by the voltage applied to the electrodes through the EO-induced spatial refractive-index modulation along the waveguide.

To analyze the characteristics of the EO mode converter, the graded-index LiNbO₃ core is modelled by an approximate equivalent step-index core that consists of two parts, as shown in Fig. 2(a). The inner part has a height h_1 , a width w_1 , and a refractive index n_1 ; the outer part has a height h_2 , a width w_2 and a refractive index n_2 . The surrounding cladding has a refractive index n_3 . The distance between the core and the buffer layer is d and the thickness of the buffer layer is t . For the simulation, we assume $n_1 = 2.161$, $n_2 = 2.143$, and $n_3 = 2.138$, $h_1 = 3.0 \mu\text{m}$, $h_2 = 6.0 \mu\text{m}$, $w_1 = 5.0 \mu\text{m}$, $w_2 = 10.0 \mu\text{m}$, and $d = 1.0 \mu\text{m}$. The Al electrodes have a thickness of 150 nm and a refractive index of $1.5237 + 15.115j$ at the wavelength 1500 nm [13]. The buffer material is SiO₂, which has a refractive index of 1.440. All the above values are chosen to match the measurement data as closely as possible. We calculate the complex effective indices and the field distributions of the modes with a commercial mode solver. The intensity distributions of the LP₀₁ and LP_{11b} modes supported by the waveguide at the wavelength 1500 nm are shown in Fig. 2(b).

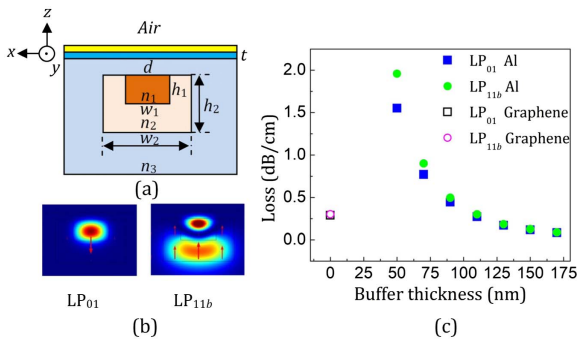


Fig. 2. (a) Cross section of the equivalent step-index waveguide used in the simulation, (b) the intensity distributions of the two guided modes supported by the waveguide, and (c) the modal losses induced by the Al electrodes with varying buffer thicknesses and by the graphene electrodes, calculated at 1500 nm.

As shown in Fig. 1(a), in the conventional electrode design, a buffer layer is placed between the waveguide and the electrodes. The thickness of the buffer layer is an important design parameter [11]. The real parts of the effective indices, which are insensitive to the buffer thickness, are 2.1521 for the LP₀₁ mode and 2.1395 for the LP_{11b} mode. The imaginary parts of the effective indices represent the metal-induced losses. The variations of the Al-induced losses of the two modes with the buffer thickness at the wavelength 1500 nm are shown in Fig. 2(c). To analyze the proposed graphene electrodes shown in Fig. 1(b), we model graphene as a conductive boundary with a complex surface conductivity of $6.084 \times 10^{-5} - 7.519 \times 10^{-6}j$ at 1500 nm [5,8] and calculate the complex effective indices of the modes. The graphene-induced losses to both modes are also shown in Fig. 2(c) (the two data points at zero buffer thickness). The graphene-induced losses are caused by the minor electric field components of the modes, which are parallel to the graphene surface. As shown in Fig. 2(c), the Al-induced losses are equal to the graphene-induced losses (0.29 dB/cm) at a buffer thickness of ~ 110 nm.

By following the method of analyzing a LiNbO₃ long-period grating [11], we calculate the transmission spectra of the EO mode converter (for a grating pitch of 119 μm and a grating length of 8.0 mm) at the wavelength 1500 nm for different applied voltages. The results for the graphene electrodes (i.e., electrodes without buffer) and the Al electrodes with a 110 nm thick buffer are shown in Figs. 3(a) and 3(b), respectively, and the variation of the half- π voltage (the voltage at which the LP₀₁ mode is completely coupled to the LP_{11b} mode) with the buffer thickness is shown in Fig. 3(c). Because the buffer layer has a refractive index much lower than that of LiNbO₃, it does not significantly affect the effective indices of the modes and, hence, the resonance wavelength. The thin electrodes also have little effect on the effective indices. Our calculation shows that the half- π voltage can be reduced by about two times by using graphene electrodes.

We should emphasize that the equivalent step-index model used in the simulation is a crude model of the fabricated LiNbO₃ waveguide; it is, however, good enough to qualitatively

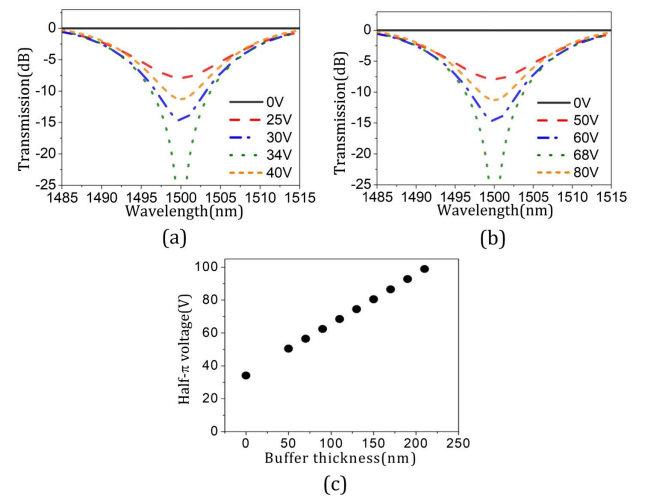


Fig. 3. Transmission spectra of the EO mode converter calculated at different applied voltages with (a) no buffer and (b) a 110 nm thick buffer. (c) Variation of the half- π voltage with the buffer thickness.

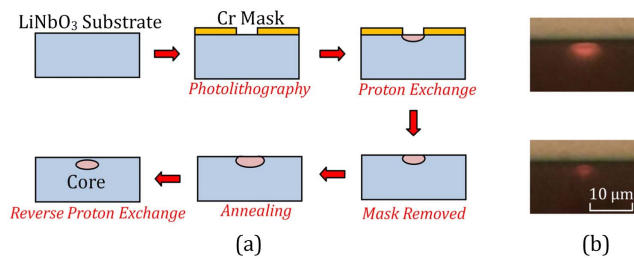


Fig. 4. (a) Steps for the fabrication of a PE LiNbO₃ waveguide and (b) images of the few-mode end (upper) and the single-mode end (lower) of a fabricated LiNbO₃ waveguide.

show the effects of replacing metal electrodes with graphene electrodes.

We fabricated the LiNbO₃ waveguide with the process illustrated in Fig. 4(a) using a *z*-cut *y*-propagation LiNbO₃ substrate. A chromium (Cr) thin film was first deposited onto the substrate, and a set of 7.0 μm wide strip openings was formed on the Cr film by the photolithography process. The sample was then immersed into melted stearic acid for 6 h at 250°C to complete the PE process, and the Cr mask was removed afterwards. To recover the EO coefficient of the LiNbO₃ waveguide after the PE process, the sample went through an annealing process followed by a reverse PE process [11,12]. The annealing was carried out in atmosphere at 350°C for 5 h, and the reverse PE was undertaken in melted lithium stearic at 300°C for 16 h. The core size of the waveguide is approximately 6.0 μm × 10.0 μm. The effective indices of the LP₀₁ and LP_{11b} modes of the waveguide, measured at 1536 nm with a prism-coupler system (Metricon 2010) [14], were 2.1521 and 2.1394, respectively. We actually fabricated a number of identical waveguides on the substrate, some of which contained a tapered core at one end. The tapered core, which had an approximate size of 3.0 μm × 5.0 μm, supported only the LP₀₁ mode. The total length of the waveguides was 15.0 mm. The images of the two end faces of a fabricated waveguide (which contained a tapered core) illuminated by white light are shown in Fig. 4(b).

According to the measured effective indices of the two modes, the grating pitch chosen was 122 μm which, according to Eq. (1), should produce a resonance at ~1550 nm. The duty cycle of the grating was 50%. We used the same fabricated LiNbO₃ waveguide to demonstrate the two electrode configurations shown in Fig. 1. Since graphene could be easily removed by swab and acetone, we first formed the graphene electrodes on the waveguide and, after characterizing the performance of the device, we removed the graphene electrodes and formed the Al electrodes, together with the SiO₂ buffer layer, on the same waveguide. The fabrication processes for the graphene electrodes and the Al electrodes are illustrated in Figs. 5(a) and 5(b), respectively. The total length of the electrodes was ~8.0 mm (66 grating periods) for both electrode configurations.

The graphene film used in this Letter (Hefei Vigon Tech.) was a mono-atom sheet with a dimension of 10 mm × 10 mm, which came as an attachment to a PMMA buffer. To transfer the graphene film onto the waveguide, the buffer was floated to deionized water and fished by the waveguide sample. The PMMA buffer was then dried and removed from the waveguide by acetone. As shown in Fig. 5(a), the transferred graphene film

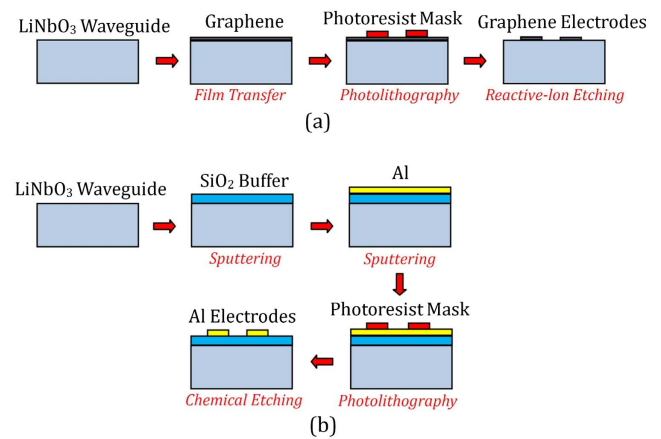


Fig. 5. Steps for the fabrication of (a) graphene electrodes and (b) Al electrodes on a LiNbO₃ waveguide.

was etched into the desired grating pattern with a photoresist mask by photolithography and reactive-ion etching (RIE) in oxygen gas. As a result, the graphene electrodes were covered with photoresist, the mask material. The excess length of the graphene film was removed by RIE. A photograph of the graphene electrodes is shown in Fig. 6(a).

To fabricate the Al electrodes, a SiO₂ layer was first sputtered onto the LiNbO₃ waveguide and etched to a thickness of 110 nm by RIE. An Al film was then sputtered onto the SiO₂ layer and etched to the desired grating pattern with a photoresist mask by photolithography and chemical etching. A photograph of the Al electrodes is shown in Fig. 6(b).

The process of fabricating the Al electrodes involves sputtering and chemical etching and requires careful control of these processes to achieve the desired quality and thicknesses for the buffer layer and the electrodes. These processes also involve environmentally harmful chemicals, such as CF₄ and CHF₃ gases. On the other hand, the process of forming graphene electrodes is intrinsically simpler, cheaper, and environment friendly. Although our process involves several manual steps, the rapid advancement in the graphene technology may well lead to an efficient process for the fabrication of graphene electrodes in the near future.

To characterize the device, we launched light from a tunable laser (HP 8168F) into the single-mode end of the waveguide with a lensed single-mode fiber (SMF) that had a mode-field diameter of 2.5 μm. The output light from the waveguide was collected with a standard G.652 SMF for power measurement. Fiber-waveguide coupling was optimized by monitoring the

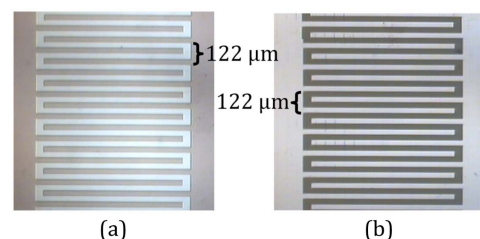


Fig. 6. Photographs of (a) graphene electrodes (with photoresist) and (b) Al electrodes formed, respectively, on the same LiNbO₃ waveguide.

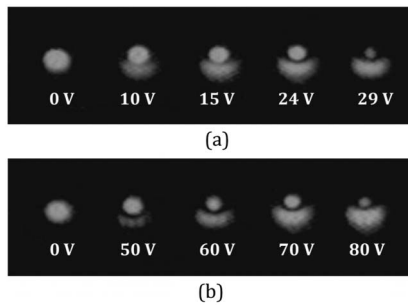


Fig. 7. Output near-field images taken at different applied voltages for (a) the graphene electrodes (at 1494 nm) and (b) the Al electrodes (at 1500 nm).

output power from the waveguide. Polarization optics was used to ensure that the light launched into the waveguide was TM-polarized. By comparing the output powers measured before and after forming electrodes on a waveguide that did not contain a tapered core, the electrode-induced losses for both graphene and Al electrodes were estimated to be 0.3 – 0.4 dB/cm at 1500 nm for both modes. As expected, for the chosen buffer thickness, the Al-induced losses were comparable to the graphene-induced losses.

To demonstrate the mode-conversion operation, we launched laser light into the single-mode end (i.e., the tapered end) of a device and captured the near-field images from the few-mode end with an infrared camera (HAMAMATSU C2714) at different applied voltages. The results for the graphene electrodes (at 1494 nm) and the Al electrodes (at 1500 nm) are shown in Figs. 7(a) and 7(b), respectively. The evolution of the output near-field pattern with the applied voltage, as shown in Fig. 7, confirms the operation of the device as an LP_{01} - LP_{11b} mode converter for both the graphene and the Al electrodes. Because the LP_{11b} mode has a highly asymmetric pattern, it is difficult to identify a pure LP_{11b} mode pattern simply from the near-field images. While it is impossible to determine the half- π voltage accurately from the near-field patterns, a comparison of the two sets of images in Fig. 7 indicates that the half- π voltage achieved with the graphene electrodes is lower than that with the Al electrodes by about three times, which agrees qualitatively with the simulation results.

Figures 8(a) and 8(b) show the normalized transmission spectra of the EO mode converter measured at different applied voltages with an optical spectrum analyzer (Agilent 86140B) at $\sim 22^\circ\text{C}$ for the graphene electrodes and the Al electrodes, respectively, when light from a broadband source (SuperK Compact) was launched into the single-mode end of the device. As shown in Fig. 8, a rejection band centered at ~ 1500 nm grows as the applied voltage increases and reaches to a maximum contrast (>20 dB) at a specific applied voltage. A further increase in the applied voltage reduces the strength of the rejection band, which is due to over-coupling (i.e., the LP_{11b} mode is coupled back to the LP_{01} mode). The slight difference between the center wavelengths of the two electrode configurations is mainly caused by the organic residues and the photoresist left on the graphene electrodes. A comparison of the two sets of the transmission spectra in Fig. 8 shows that the half- π voltage achieved with the graphene electrodes is lower than that with the Al electrodes by almost three times. The Al electrodes

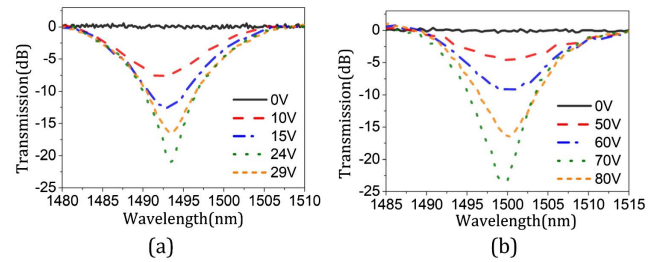


Fig. 8. Transmission spectra of the experimental EO mode converter measured at different applied voltages at $\sim 22^\circ\text{C}$ for (a) the graphene electrodes and (b) the Al electrodes, when only the LP_{01} mode was launched into the device.

would have to be almost three times longer to achieve the same half- π voltage achieved with the graphene electrodes, and the resultant bandwidth would be much narrower. We should note that the fiber-waveguide coupling at the output end of the waveguide was optimized to produce the largest possible rejection band. We also measured the temperature sensitivity of the transmission spectrum of the device. The rejection band shifted towards the longer wavelength at $1.9 \text{ nm}/^\circ\text{C}$ for both electrode configurations.

In conclusion, we have proposed and demonstrated the idea of using graphene electrodes for EO devices with an EO-grating mode converter fabricated on a z -cut LiNbO_3 substrate. The use of graphene electrodes can do away with the buffer layer and, thus, reduce the half- π voltage of the device. For our experimental device, the half- π voltage achieved with the graphene electrodes is lower than that achieved with the conventional Al electrodes by almost three times. With the buffer layer exempted, the process of fabricating the EO device is also simplified. The use of graphene electrodes can significantly enhance the efficiency of LiNbO_3 EO devices and potentially reduce their fabrication cost. While this Letter focusses on LiNbO_3 , it is obvious that graphene electrodes can be applied to other EO materials.

Funding. Research Grants Council, University Grants Committee (RGC, UGC), Hong Kong (CityU 11202014, CityU 11205715).

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